

Supplementary information for

Detachable, self-repairing and closed-loop recyclable self-initiated light-curing adhesives with controllable bonding strength

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Characterization and methods

1.1 Structural characterization and crystallization behavior of CPTA

The nuclear magnetic resonance (NMR) spectrum was performed on Bruker Avance III-400 NMR spectrometer with DMSO-d₆ as the solvent and TMS as the internal reference. MALDI-TOF mass spectrometry analysis was carried out on a Bruker-autoflex speed from Germany with the matrix of DCTB (trans-2-[3-(4-tertbutylphenyl)-2-methyl-2-propenylidene]-malononitrile). The UV-vis spectra and light transmittance were characterized by TU-1901 UV-vis spectrophotometer from 230 nm to 500 nm. Fourier transform infrared spectroscopy (FTIR) was measured by Bruker Tensor 27 spectrometer in transmission mode with the wavenumber range of 4000 - 400 cm⁻¹. The viscosity was carried out on Rotary Rheometer (TA DHR-20) by using parallel plates with a diameter of 20 mm, control plates with a spacing of 1000 μm and vibration shear deformation mode. The infrared images were obtained using an infrared thermal imager (Fluke Ti450). The XPS spectra was tested in Kratos AXIS Ultra DLD X-ray Photoelectron Spectroscopy. The XRD spectra were analyzed in an X-ray diffractometer (D8 Advance, Bruker). The POM images were recorded by polarizing microscope (Motic BA310Pol).

The thermal transition test was obtained on Mettler-Toledo differential scanning calorimeter DSC, and the heating rate was 10 °C/min in nitrogen atmosphere. The thermal history of the material is eliminated by heating and cooling cycles, and the second heating curve is taken as the DSC curve of the material. The enthalpy of

crystallization melting (ΔH) is obtained by integrating the peak area of DSC curve. The calculation formula of enthalpy change is:

$$\Delta H = \int Q dt$$

Where ΔH is enthalpy change, Q is heat flow and t is time. The measured enthalpy is expressed in joules per gram (J/g). Taking the crystallization enthalpy of fusion of CPTA as 100% crystallinity, the crystallinity of other samples is determined by the percentage of their crystallization enthalpy of fusion relative to that of CPTA.^[1]

Thermogravimetric analysis (TG) was determined by Mettler Toledo TGA, and the heating rate was 10 °C/min in nitrogen atmosphere. The hydrophilicity and hydrophobicity of materials were tested in PT-705C contact angle measuring instrument.

1.2 The adhesive and detachment behavior of CPTA

The adhesive prepolymer was melted at 80 °C, and then quickly and evenly coated on the surface of glass block washed with alcohol. Covering another glass block and press it gently to immerse the adhesive. After waiting for different time, the samples were irradiated with UV light (intensity of 50 mW/cm²) for 2 min to obtain adhesives with different crystallinity. The maximum shear strength was conducted on compression mode in CMT 6303 electronic universal testing machine. Each sample was tested at least three times with a loading rate of 1 mm/min. The tensile properties were collected on AGS-X instrument equipped with a 50 N load cell. Each sample was tested at least three times with a loading rate of 2 mm/min. Scanning electron

microscope (SEM) images were observed on a scanning electron microscope (Zeiss 73477).

The energy of interfacial adhesive and cohesive:

To explore the adhesion behaviors of **CPTA** with glass, the interaction mechanisms were analyzed at the atomic level by molecular dynamics (MD) simulations. MD simulations were performed on polymer/SiO₂ MXene system using Gromacs 2019.4 package.^[2-3] The initial structure of the polymer was constructed using *GaussView 6*,^[4] followed by structural optimization via *Gaussian 16*^[5] to obtain the corresponding electrostatic potential distribution. Finally, the RESP charge distribution of the polymer system was calculated using Multiwfn 3.8_dev.^[6] The force field parameters for polymer and SiO₂ were obtained through SobTop software.^[7] The size of the glass is 8.65 nm × 8.65 nm × 1.90 nm. And then 61 polymer chains were placed on the surface of the glass. To get the equilibrium structure of the simulation system, 10 ns NVT equilibrium simulation were performed firstly. Then, 10 ns NPT runs were performed respectively at 298 K and 353K to obtain adhesion energy and cohesion energy. The adhesion energy and cohesive energy are calculated respectively by using the following formulas: $E_{adhesion\ energy} = E_{tot} - E_{SiO_2} - E_{polymers}$, $E_{cohesive\ energy} = E_{polymers} - E_{single\ polymer}$. Herein, E_{tot} is the total energy of the simulation system, E_{SiO_2} is the energy of SiO₂ substate, $E_{polymers}$ is the total energy of polymers, and $E_{single\ polymer}$ is the energy of single polymer. The visualization of the simulation structures is made by using VMD software.^[8]

Dimensional change was measured by Thermomechanical Analysis (TMA) on TA TMA450. Samples were heated from 25 to 125 °C at a heating rate of 5 °C/min with a loading force of 0.2 N in nitrogen atmosphere. The inversion spectrum of transverse relaxation time T_2 of NMR was analyzed by using a low-field NMR analyzer (PQ001). The cured sample was cut 5mm×30mm rectangular strips directly into a specific nuclear magnetic tube for scanning test, and the horizontal coordinate of the obtained spectrum represented the relaxation time of the sample. The stress relaxation and creep were performed by TA DMA850. The stress relaxation samples were heated at 25 and 120 °C with 4% strain for 30 min. The creep samples were heated at 25 and 120 °C with 0.1 MPa stress for 20 min.

1.3 Degradation and closed-loop recycle behavior of PCPTA-*n*

Raman spectroscopy was performed on Raman spectrometer (WITec Alpha300R, 532 nm) with the Raman shift range of 300 - 2800 cm^{-1} . The cytotoxicity assay in vitro was carried out in the following ways. **PCPTA-0** and **PCPTA-20** films were directly cut into 3mm×3mm squares for experiments. TPGDA was selected as the photoactive monomer, and a photoinitiator with a mass fraction of 1% was added for UV curing for 5min, and then cut into 3mm×3mm squares for use. A375 cells was seeded in 96-well plates at an initial cell density of 1.0×10^4 cells per well and incubated overnight. Then, the medium was replaced by 100 μL fresh medium with different adhesives film and incubated for 24 h. Finally, the culture medium was replaced with 100 μL 5% CCK-8 solution and incubated at 37 °C for 1 h. Gently shaking the incubated culture medium to make the substances in the pores evenly distributed in the pores. Cell viabilities were

quantified by measuring the absorbance at 450 nm and normalized to the control group.

The cell viability was calculated by the following formula

$$\text{The cell viability} = \text{OD}_{\text{sample}} / \text{OD}_{\text{control group}} \times 100\%$$

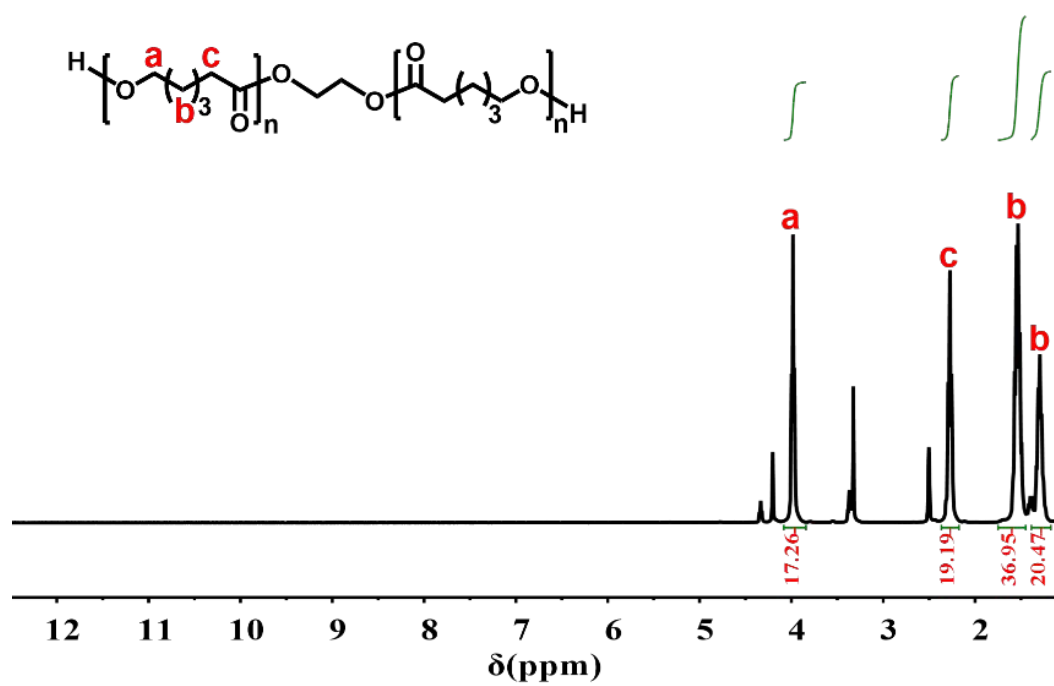


Figure S1 ^1H NMR spectrum of PCL in DMSO-d_6 .

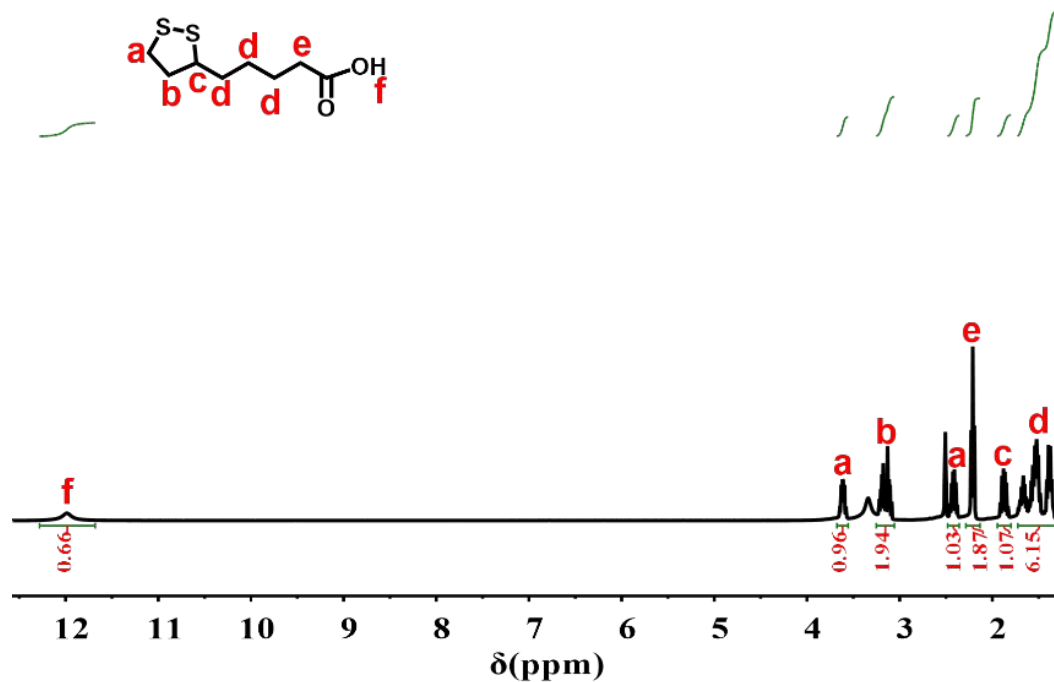


Figure S2 ^1H NMR spectrum of TA in DMSO-d_6 .

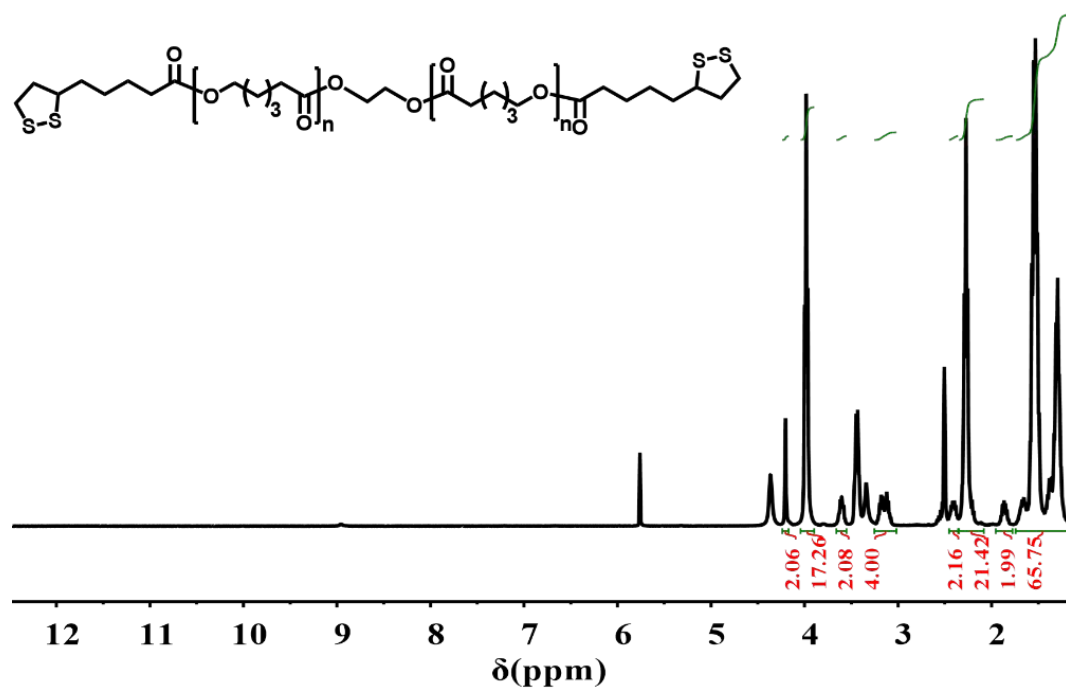


Figure S3 ^1H NMR spectrum of CPTA in DMSO-d_6 .

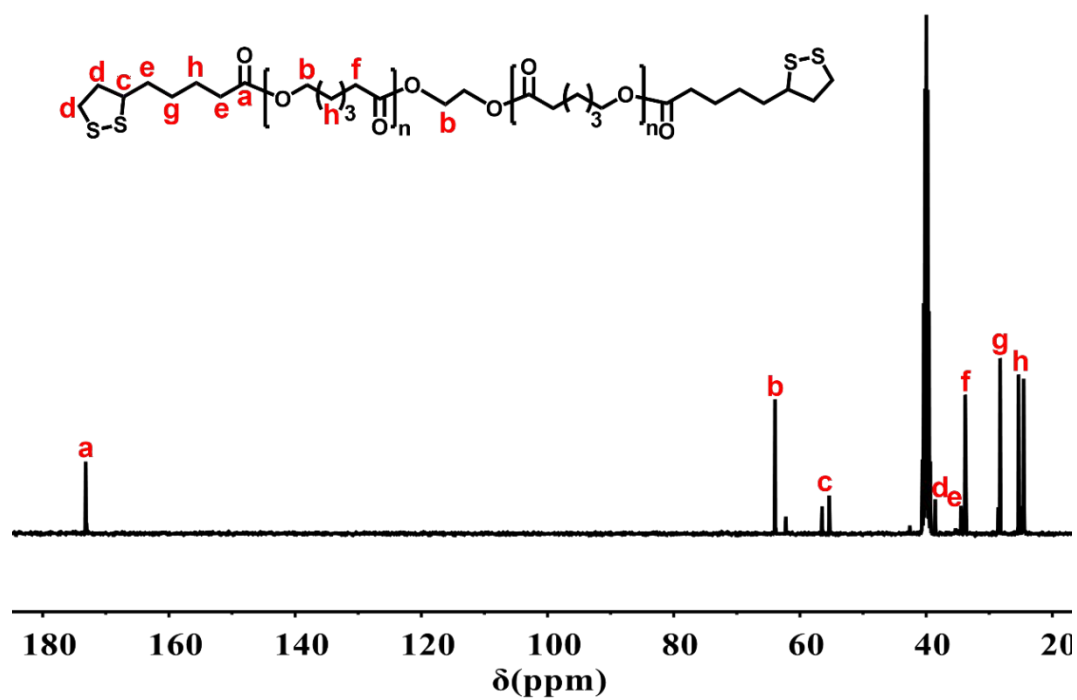


Figure S4 ^{13}C NMR spectrum of CPTA in DMSO-d_6 .

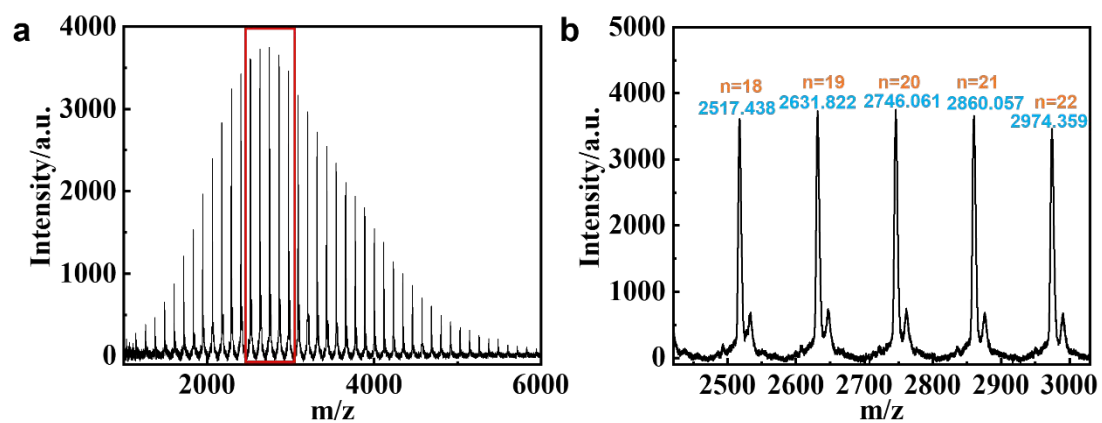


Figure S5 (a) MALDI-TOF mass spectrum of **CPTA**. (b) Partial MALDI-TOF mass spectrum of **CPTA**.

Table S1 Results of MALDI-TOF mass spectrum analysis of **CPTA**.

Polymerization degree	18	19	20	21	22
Measured M_w	2517.438	2631.822	2746.061	2860.057	2974.359
Theoretical M_w	2517.72	2631.95	2746.18	2860.41	2974.64

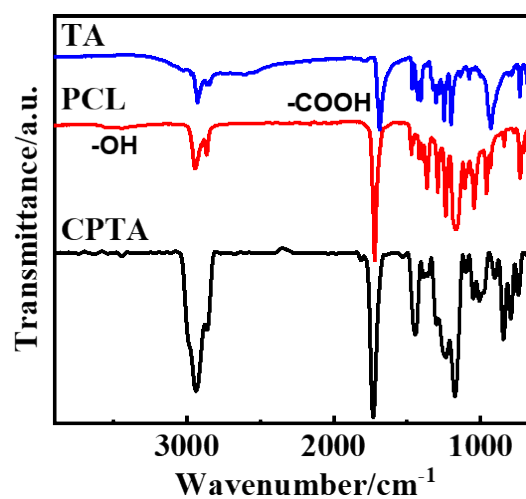


Figure S6 The FTIR spectra of PCL, TA and CPTA.

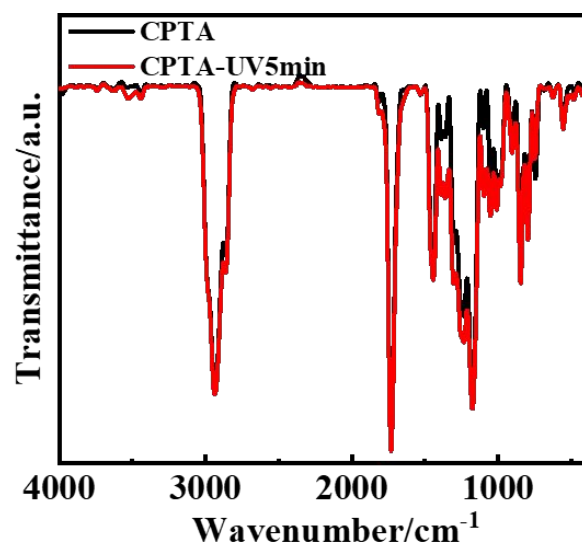


Figure S7 The FTIR spectra of CPTA and irradiation under UV light for 5 min.

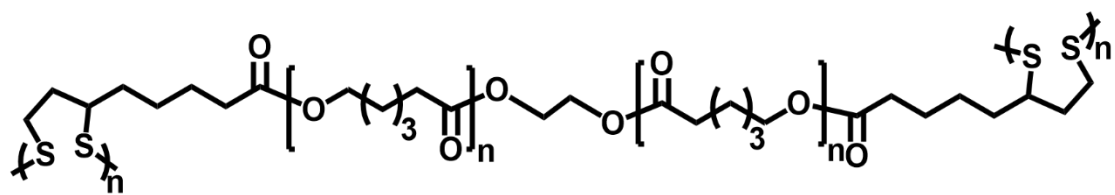


Figure S8 The chemical structure of **PCPTA**.

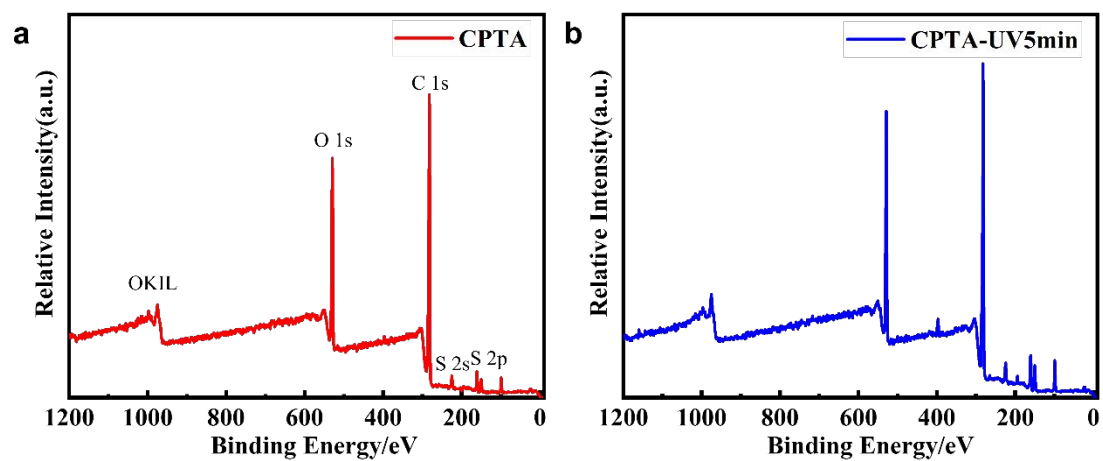


Figure S9 The XPS spectra of (a) CPTA and (b) irradiation under UV light for 5 min.

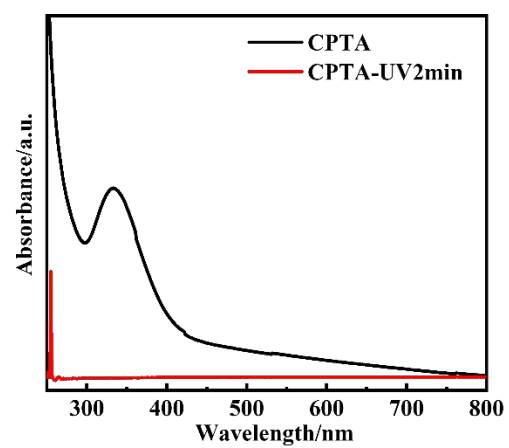


Figure S10 UV-vis spectra of CPTA and irradiation under UV light for 2 min.

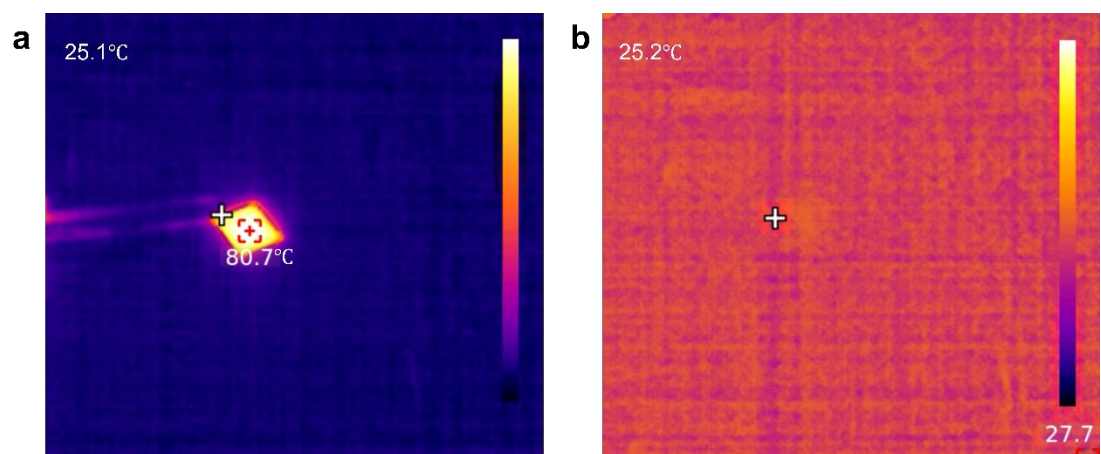


Figure S11 Comparison of infrared images between CPTA at different temperatures and 25 °C background (a) at 80 °C, (b) at 25 °C.

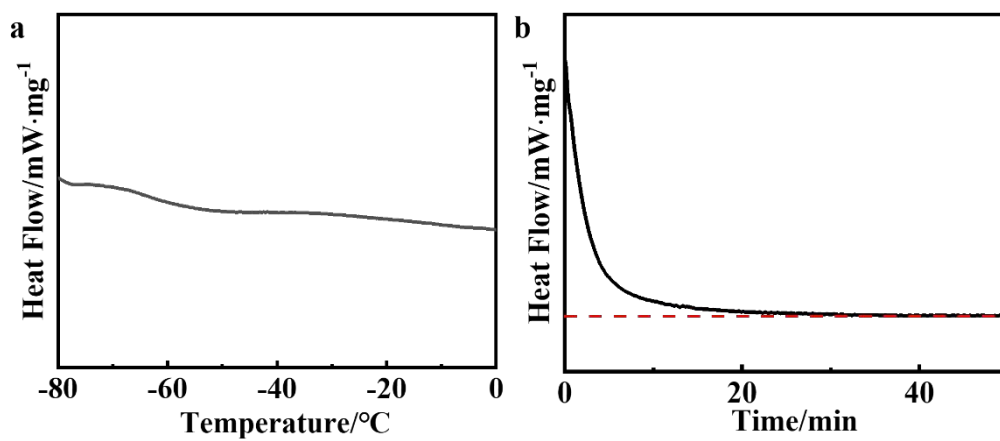


Figure S12 (a) DSC curve of CPTA with temperature change. (b) DSC curve of molten CPTA with time at room temperature.

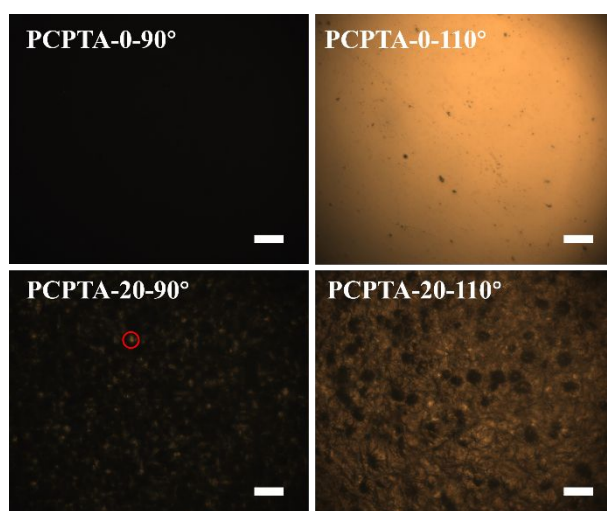


Figure S13 The polarized light microscope images of **PCPTA-0** and **PCPTA-20** (scale bar=100 μm). (When the main refractive index direction of the crystal is consistent with the direction of polarized light, the light will not be deflected, so it will remain dark (extinction position). However, when the orientation of the crystal changes, part of the light can pass through the analyzer, resulting in a change in brightness.)

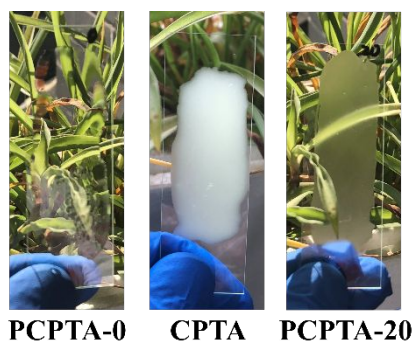


Figure S14 The physical pictures of **PCPTA-0**, **CPTA** and **PCPTA-20**.

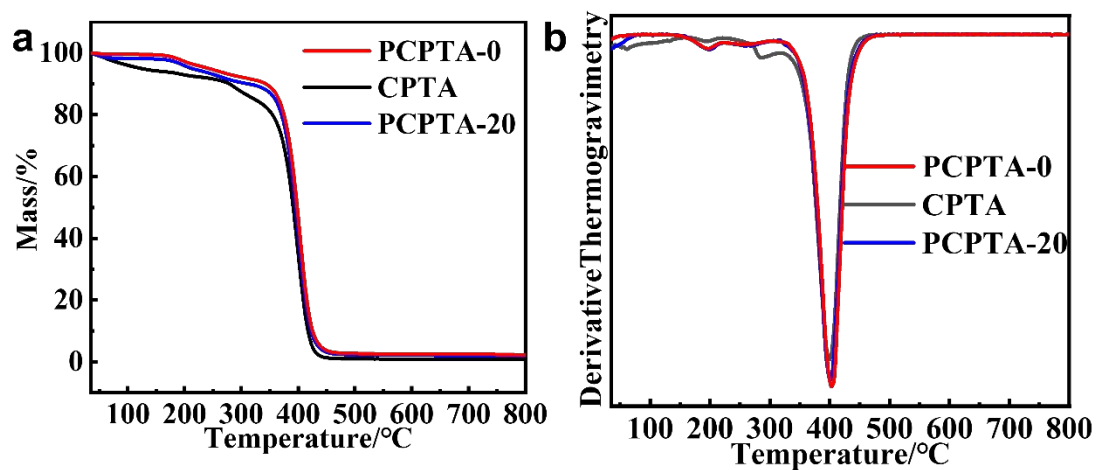


Figure S15 (a) The TG curves and (b) the DTG curves of PCPTA-0, CPTA and PCPTA-20.

Table S2 Thermal properties of LCA.

Sample	PCPTA-0	CPTA	PCPTA-20
$T_{5\%}$ (°C)	212.5	123.3	242.2
$T_{\max}$ (°C)	403.4	399.6	401.2

Table S3 Crystallization enthalpy and degree of crystallinity of LCA.

Sample	CPTA	PCPTA-5	PCPTA-10	PCPTA-15	PCPTA-20
Crystallization enthalpy (J/g)	73.20	14.27	25.65	38.48	51.57
Degree of crystallinity (%)	100	19.5	35.0	52.6	70.4

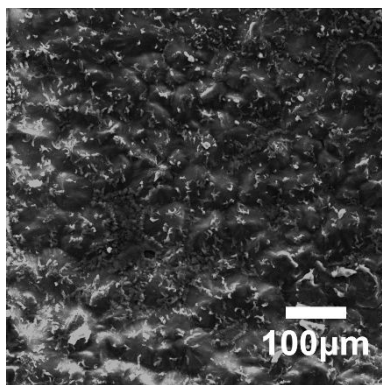


Figure S16 The SEM picture of **PCPTA-20** surface.

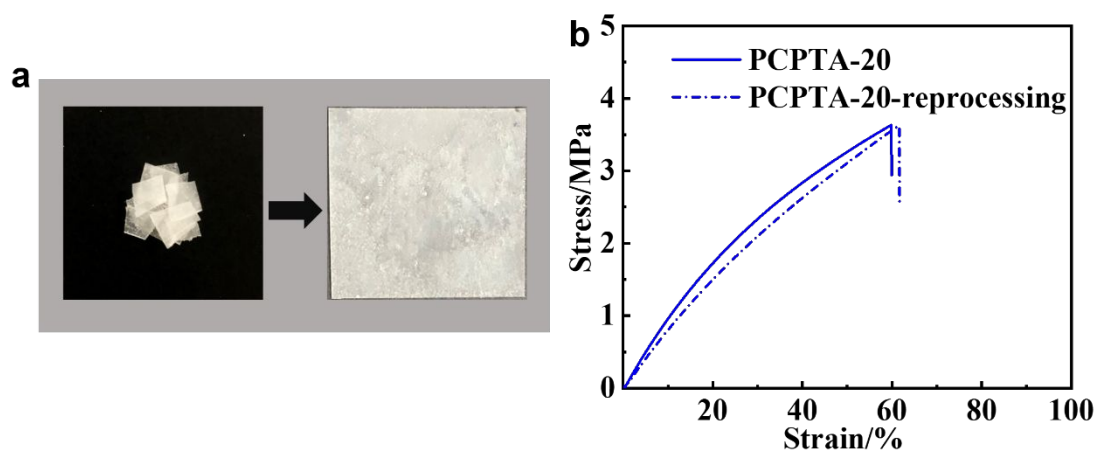


Figure S17 (a) Remolding process of **PCPTA-20** heated at 70 °C for 1h. (b) Stress-strain curves of reprocessed **PCPTA-20**.

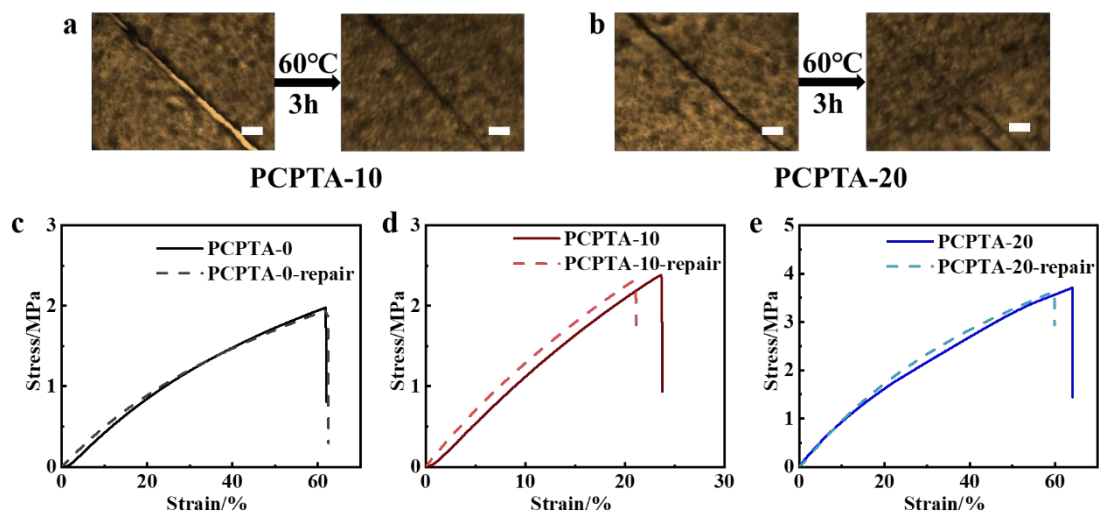


Figure S18 Optical microscopic images of a scar on the film and healing process for (a) **PCPTA-10** and (b) **PCPTA-20** (scale bar=200 μm). The stress-strain curves of (c) **PCPTA-0**, (d) **PCPTA-10** and (e) **PCPTA-20** before and after repair.

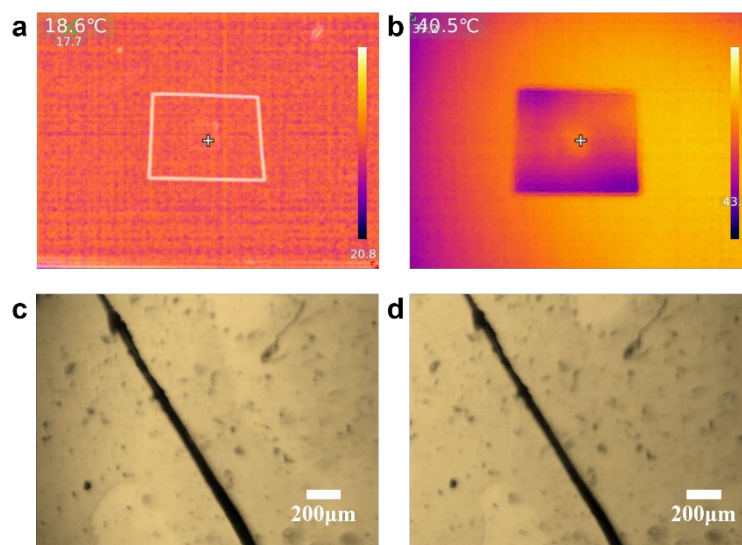


Figure S19 Infrared images of **PCPTA-20** (a) at room temperature, (b) after 10 min of UV radiation. Optical microscopic images of a scar on the film and healing process for **PCPTA-20** in (c) 25 °C and (d) 45 °C.

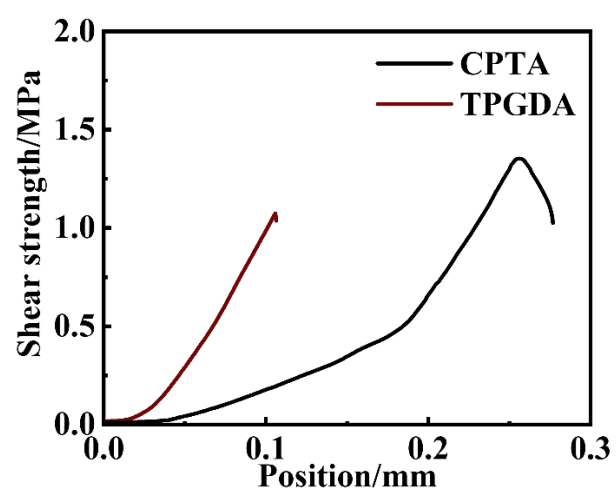


Figure S20 The shear strength curves of CPTA and TPGDA.

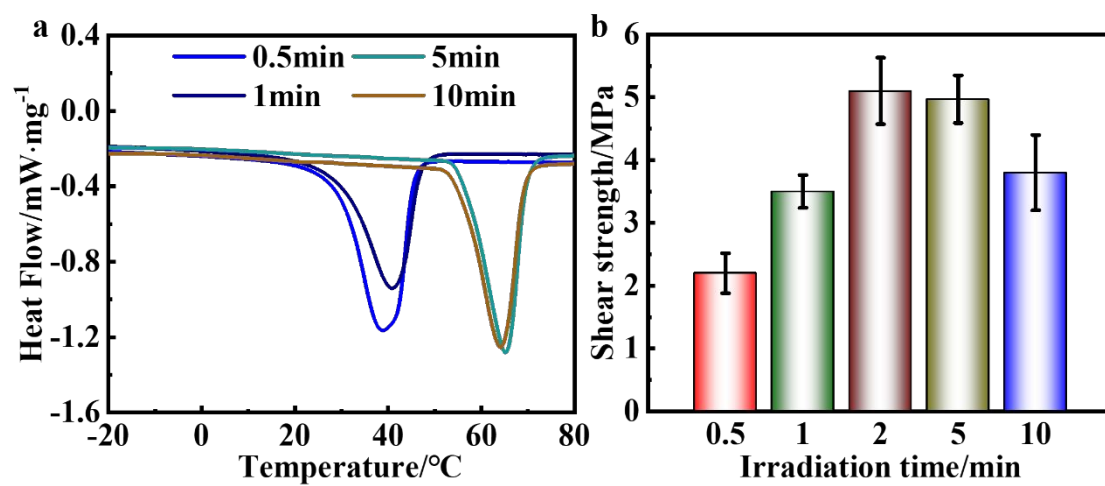


Figure S21 (a) DSC curves and (b) Shear strength of **PCPTA-20** with the extension of irradiation time.

Table S4 Crystallization enthalpy and degree of crystallinity of **PCPTA-20** with the extension of irradiation time.

Time (min)	0.5	1	2	5	10
Crystallization enthalpy (J/g)	64.17	53.46	51.57	50.80	49.78
Degree of crystallinity (%)	87.7	73.1	70.4	69.4	68.0

Table S5 Mechanical properties of LCA.

Sample	PCPTA-0	PCPTA-5	PCPTA-10	PCPTA-15	PCPTA-20
Shear-strength (MPa)	1.3	1.9	4.1	4.69	5.1
Tensile-strength (MPa)	1.9	2.3	2.4	3.3	3.6

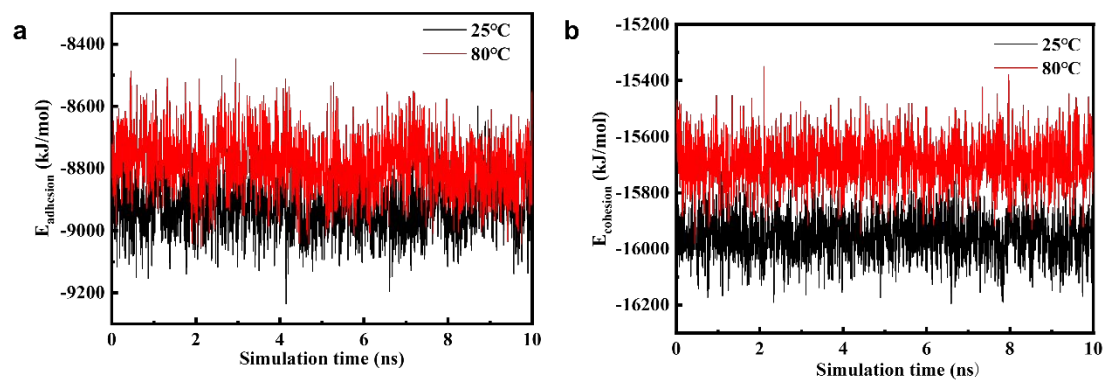


Figure S22 Variation curves of E_{cohesion} and E_{adhesion} of CPTA with time at (a) 25 °C (b) and 80 °C.

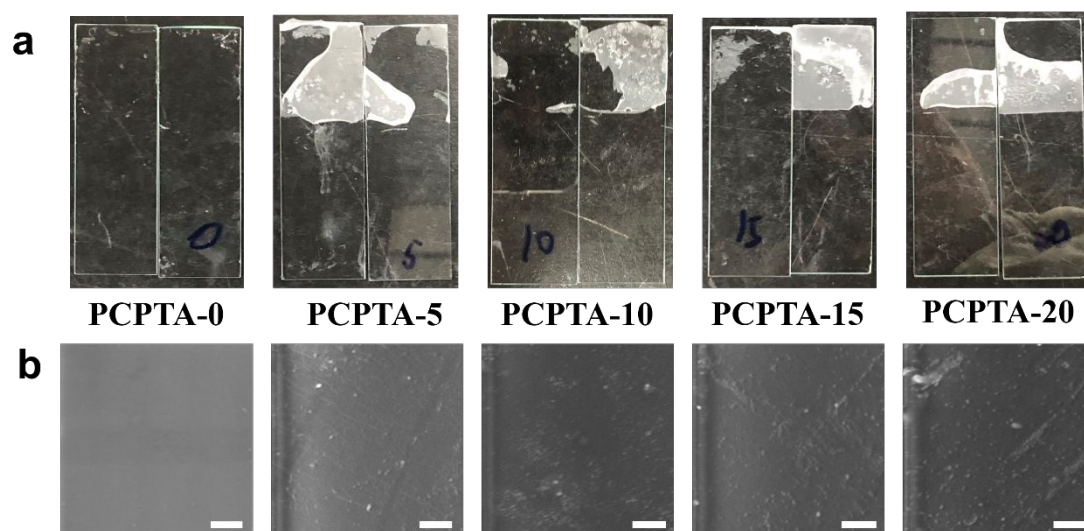


Figure S23 (a) The physical and (b) SEM pictures of the detachment of **PCPTA-*n***.

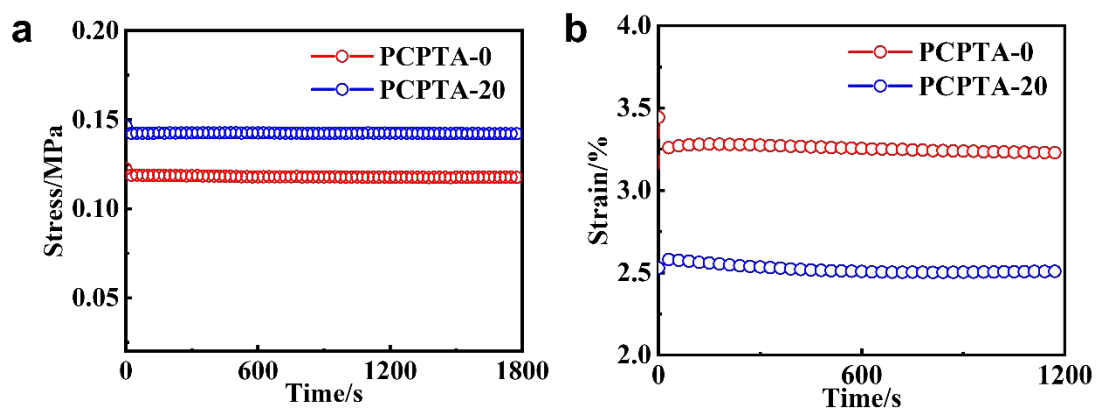


Figure S24 (a) The stress relaxation and (b) creep of **PCPTA-0** and **PCPTA-20** at 25 °C.

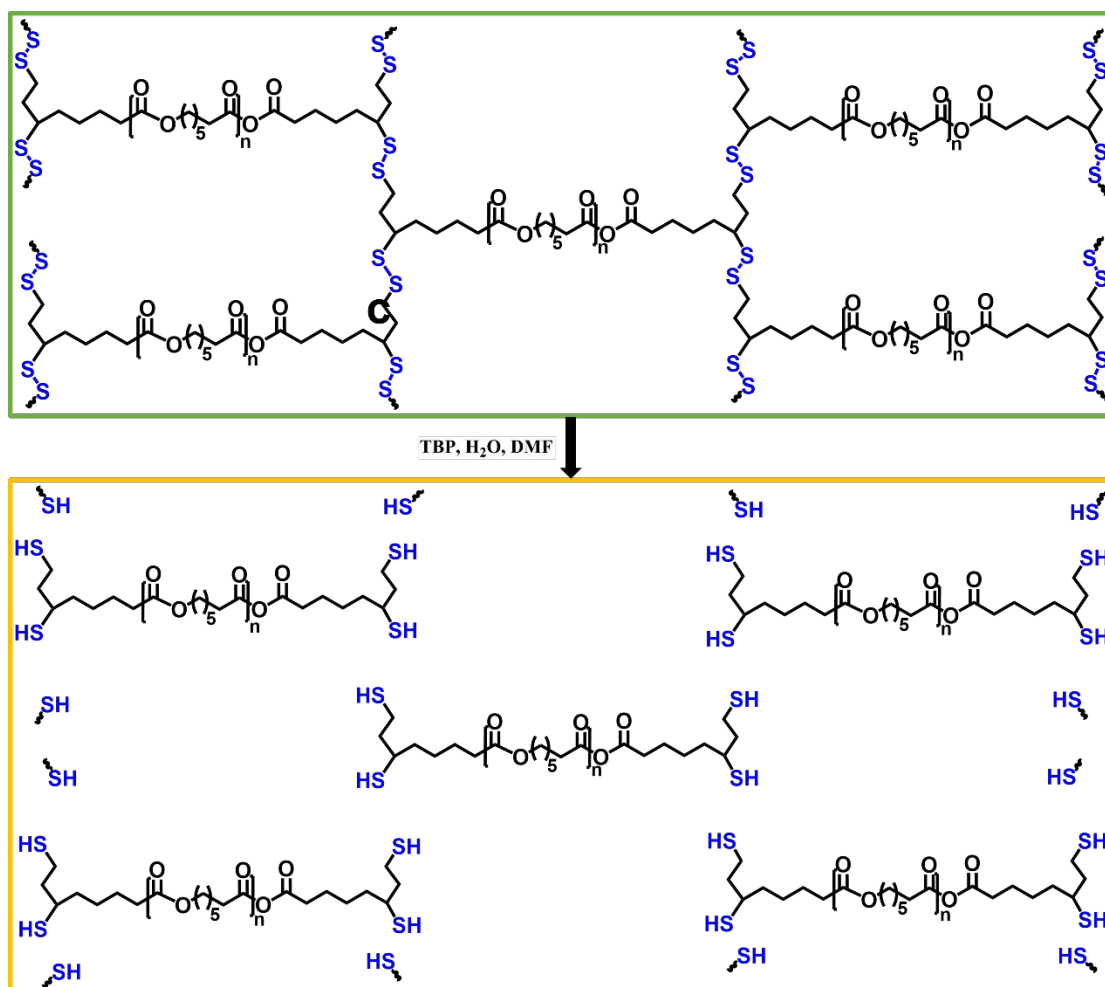


Figure S25 Mechanism diagram of degradation process.

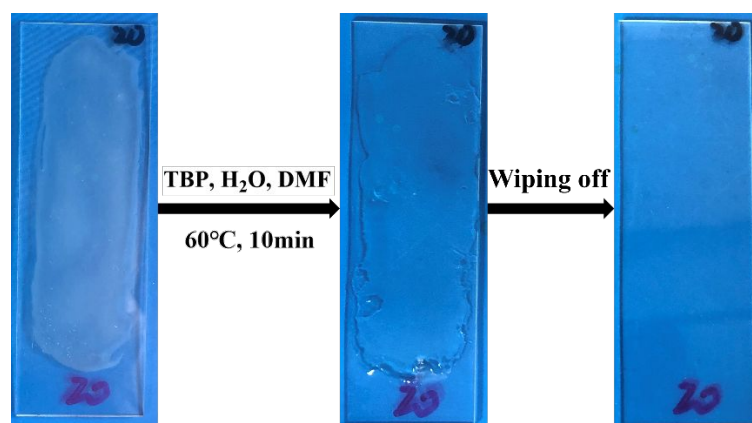


Figure S26 Removable test of LCA.

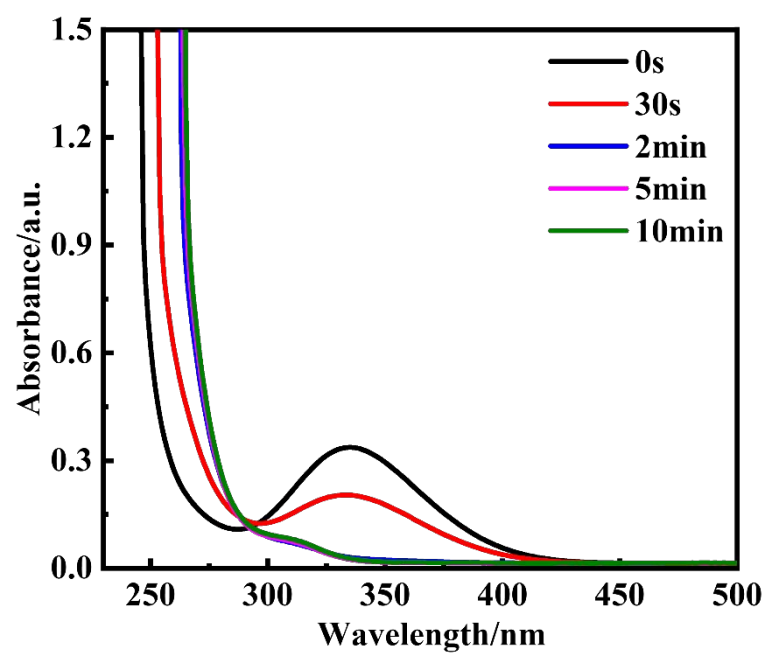


Figure S27 UV-vis spectra of recycled CPTA (1×10^{-3} mol/L) in dichloromethane following 365 nm light irradiation.

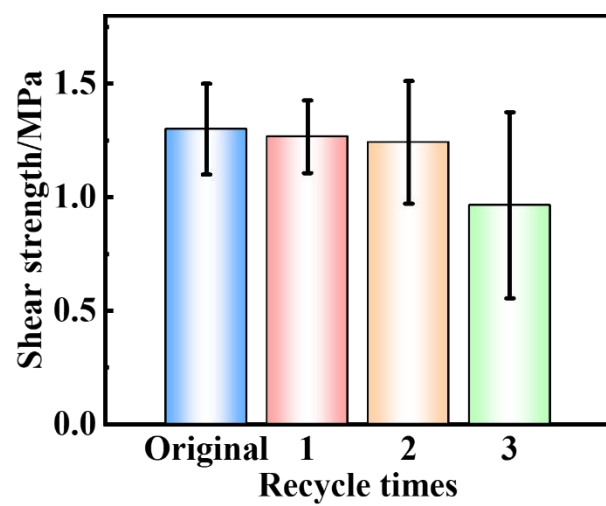


Figure S28 The shear strength of PCPTA-0 with different recycle times.

Table S6 Comparison of **PCPTA** with other adhesives.

Number	Light-curing	Bonding strength (MPa)	Recyclable	Detachment
Our work	√	5.1	√	√
Ref. ^[9]	√	2.0	√	√
Ref. ^[10]	√	1.1	√	√
Ref. ^[11]	√	3.5	×	√
Ref. ^[12]	×	5.3	√	√
Ref. ^[13]	×	4.2	√	√
Ref. ^[14]	×	2.2	√	√
Ref. ^[15]	×	7.0	√	√
Ref. ^[16]	×	0.6	√	√
Ref. ^[17]	×	7.3	√	√
Ref. ^[18]	×	24.4	√	√
Ref. ^[19]	×	0.09	√	√
Ref. ^[20]	×	18.6	√	√
Ref. ^[21]	×	3.1	√	√
Ref. ^[22]	×	41.0	√	√
Ref. ^[23]	×	3.5	√	√
Ref. ^[24]	√	8.6	×	×
Ref. ^[25]	√	7.6	×	×
Ref. ^[26]	√	4.8	×	×
Ref. ^[27]	×	8.0	×	√

Ref. ^[28]	×	10.9	×	×
Ref. ^[29]	×	21.0	×	×
Ref. ^[30]	×	20.7	×	×
Ref. ^[31]	×	5.0	×	×
Ref. ^[32]	×	18	×	×

References

- [1] J Wang, H Zhang, J. J. Lei, M. X. Wu, W. F. Liu and J. P. Qu, *Macromolecules* **2022**, *55*, 10113-10123.
- [2] M. J. Abraham, T. Murtola, R. Schulz, S. Páll, J. C. Smith, B. Hess, E. Lindahl, *SoftwareX*, **2015**, 1-2, 19-25.
- [3] D. Van Der Spoel, E. Lindahl, B. Hess, G. Groenhof, A. E. Mark, H. J. C. Berendsen, *J. Comput. Chem.* , **2005**, *26*, 1701-1708.
- [4] GaussView, Version 6, Roy Dennington, Todd A. Keith, and John M. Millam, Semichem Inc., Shawnee Mission, KS, 2016.
- [5] Gaussian 16, Revision B.01, M. J. Frisch, G. W. Trucks, H. B. Schlegel, et al., Gaussian, Inc., Wallingford CT, 2016.
- [6] Lu, T.; Chen, F., Multiwfn: a Multifunctional Wavefunction Analyzer. *J. Comput. Chem.* 2012, *33*, 580-592.
- [7] T. Lu, Sobtop, Version 1.0, <http://sobereva.com/soft/Sobtop>
- [8] W. Humphrey, A. Dalke, K. Schulten, *J. Mol. Graphics*, **1996**, *14*, 33.
- [9] S. Ito, H. Akiyama, R. Sekizawa, M. Mori, M. Yoshida and H. Kihara, *ACS Appl. Mater. Interfaces*, **2018**, *10*, 32649.
- [10] X. Huang, Z. Shanguan, Z.-Y. Zhang, C. Yu, Y. He, D. Fang, W. Sun, Y.-C. Li, C. Yuan, S. Wu and T. Li, *Chem. Mater.*, **2022**, *34*, 2636.
- [11] D. Kim, H. Kim, W. Jeon, H. J. Kim, J. Choi, Y. Kim and M. S. Kwon, *Adv. Mater.*, **2024**, *36*, 2309891.
- [12] B. T. Michal, E. J. Spencer and S. J. Rowan, *ACS Appl. Mater. Interfaces*, **2016**, *8*, 11041.

- [13] Z. H. Zhao, P. C. Zhao, Y. Zhao, J. L. Zuo and C. H. Li, *Adv. Funct. Mater.*, **2022**, 32, 2201959.
- [14] H. Chen, X. Liu, Q. He, S. Zhang, S. Xu and Y. Z. Wang, *Adv. Mater.*, **2023**, 36, 2310779.
- [15] F. Van Lijsebetten, T. Maiheu, J. M. Winne and F. E. Du Prez, *Adv. Mater.*, **2023**, 35, 2300802.
- [16] B. S. Pi, K. Lee, C. Song, B. Jeon, Y. Han and S. Q. Choi, *Chem. Eng. J.*, **2024**, 495, 153392.
- [17] X. Ou, X. Zou, Q. Liu, L. Li, S. Li, Y. Cui, Y. Zhou and F. Yan, *Chem. Mater.*, **2023**, 35, 1218.
- [18] J. Xiong, M. Duan, X. Zou, S. Gao, J. Guo, X. Wang, Q. Li, W. Li, X. Wang and F. Yan, *J. Am. Chem. Soc.*, **2024**, 146, 13903.
- [19] Y. Zhang, X. Li, G. Bai, W. Wei and X. Liu, *J. Mater. Chem. B*, **2021**, 9, 5818.
- [20] C. Bao, Y. Yin, Y. Ding, J. Liu, Y. Xu, R. Miao, Z. Liu, B. Duan, Y. Qin and Z. Xin, *Macromolecules*, **2023**, 56, 6633.
- [21] X. Xie, Y. Jiang, X. Yao, J. Zhang, Z. Zhang, T. Huang, R. Li, Y. Chen, S.-L. Li and Y.-Q. Lan, *Nat. Commun.*, **2024**, 15, 5017.
- [22] M. A. Rahman, C. Bowland, S. Ge, S. R. Acharya, S. Kim, V. R. Cooper, X. C. Chen, S. Irle, A. P. Sokolov, A. Savara, T. Saito, *Sci. Adv.*, **2021**, 7, eabk2451
- [23] Z.-H. Wang, B.-W. Liu, F.-R. Zeng, X.-C. Lin, J.-Y. Zhang, X.-L. Wang, Y.-Z. Wang and H.-B. Zhao, *Sci. Adv.*, **2022**, 8, eadd8527.
- [24] J. H. Back, Y. Kwon, H. Cho, H. Lee, D. Ahn, H. J. Kim, Y. Yu, Y. Kim, W. Lee

and M. S. Kwon, *Adv. Mater.*, **2022**, *35*, 2204776.

[25] Y. Zhou, C. Zhang, S. Gao, B. Zhang, J. Sun, J.-j. Kai, B. Wang and Z. Wang, *Chem. Mater.*, **2021**, *33*, 8822.

[26] Y. Deng, Z. Huang, B. L. Feringa, H. Tian, Q. Zhang and D.-H. Qu, *Nat. Commun.*, **2024**, *15*, 3855.

[27] Y. Kwon, S. Lee, J. Kim, J. Jun, W. Jeon, Y. Park, H.-J. Kim, J. Gierschner, J. Lee, Y. Kim and M. S. Kwon, *Nat. Commun.*, **2024**, *15*, 2829.

[28] X. Ou, J. Pan, Q. Liu, Y. Niu, Y. Zhou and F. Yan, *Adv. Mater.*, **2024**, *36*, 2312906.

[29] P. Niu, C. Li, J. Zhu, Y. Zhao, Z. Li, A. Sun, L. Wei, K. Wu and Y. Li, *Angew. Chem. Int. Edit.*, **2024**, e202408840.

[30] P. Zhang, B. Zhang, J. Pan, G. Zhang, C. Ma and G. Zhang, *Chem. Mater.*, **2023**, *35*, 7730.

[31] Z. Dong, J. Wu, X. Shen, Z. Hua and G. Liu, *Chem. Sci.*, **2023**, *14*, 3938.

[32] C. R. Westerman, B. C. McGill and J. J. Wilker, *Nature*, **2023**, *621*, 306.